Voice & Conversation Capability Specification

Project NOVA Voice Interaction, Speech Transcription, and Conversation Management

---

Field	Value
Document ID	NOVA-SPEC-008
Version	1.0
Status	`APPROVED`
Author	Antigravity (Lead Software Engineering Agent)
Reviewer	ChatGPT (Chief Architect)
Approved By	Praveen (Project Founder)
Created	2026-06-28
Last Updated	2026-06-28
Dependencies	NOVA-SPEC-001, NOVA-SPEC-002, NOVA-SPEC-003, NOVA-SPEC-004

Revision History

Version	Date	Author	Summary of Changes
1.0	2026-06-28	Antigravity	Consolidate Voice Overview, Wake Word, STT, TTS, Conversation Management, and Provider Abstraction

specifications.

Table of Contents

Purpose & Scope

This specification defines the functional boundaries, API contracts, and safety policies for the **Voice & Conversation Capability**. It enables NOVA to receive audio signals, identify wake triggers, transcribe speech, generate voice playbacks, and track conversational session contexts.

Constraint: Voice acts as a communication interface. Conversational planning and intention analysis are handled by the `AIKernel` and `PlannerEngine`.

Wake Word Detection & Triggers

- **Wake Word:** Listens to continuous audio feeds to match configured hotwords (e.g. *"Hey Nova"*).
- **Resource Bounds:** Wake word routines must execute locally with minimal CPU loads.
- **Push-to-Talk:** Exposes trigger methods to manually initiate recording sessions without wake-word monitoring.

Speech-to-Text (STT) Subsystem

- **Audio Transcription:** Converts incoming microphone signal buffers into text strings.
- **Streaming Outputs:** Emits transcribed tokens in real-time before sentences complete.
- **Confidence Metrics:** Assigns probability ratings to transcribed tokens.

Text-to-Speech (TTS) Subsystem

- **Speech Generation:** Converts output response strings into natural audio buffers.
- **Voice Profiles:** Supports configurable gender, style, and tone profiles.
- **Interruptible Playback:** Exposes abort hooks to instantly stop audio hardware playback channels.

Conversation Manager & Session Context

- **Session History:** Records chat text logs and context tokens.
- **Context Windowing:** Manages conversational limits by pruning or summarizing oldest logs.
- **Follow-Up Trees:** Handles unresolved questions and multi-turn loops.

Provider Abstraction Interfaces

To ensure model independence, the capability isolates audio engines behind two interface classes:

```
class ISTTProvider(ABC):
    @abstractmethod
    def transcribe_stream(self, audio_generator: Generator[bytes, None, None]) -> Generator[str, None, None]: ...
    @abstractmethod
    def transcribe_file(self, file_path: str) -> dict: ...

class ITTSProvider(ABC):
    @abstractmethod
    def synthesize_speech(self, text: str, voice_profile: str) -> bytes: ...
```

Examples: Whisper, Vosk, Piper, Coqui, or Azure Speech APIs implement these interfaces.

Latency, Privacy & Interruption Policies

9. **Conversational Turnaround Latency:** Inbound user audio to outbound TTS playback start latency must be under **1.2 seconds**. To satisfy this, transcription must stream blocks in real-time, allowing intent detection to run before sentences terminate.
10. **Visual Indicators:** A hardware-independent on-screen indicator (e.g., status tray icon) must highlight when the microphone is recording.
11. **Speaker Interruption Detection:** If user speech is registered during speaker playback, the system must trigger a `PlaybackInterrupted` event, stop synthesis playback, and toggle back to input recording.

Testing & Noise Resilience Targets

- **Wake-Word Performance:** False positives must be under 1 per 24 hours of ambient room noises.
- **STT Word Error Rate (WER):** Target WER must be under 10% under standard office background noises.
- **Interruption Latency:** Synthesis abort turnaround must take less than 150ms after speech is registered.